

ArabGN-CF

System Architecture and Design Specification

Field	Value
Project	Name Redaction Does Not Blind an Arabic Résumé Ranker
Target venue	NILES 2026, Special Session SS2
Document status	Design specification, v0.1
Last updated	12 August 2026

1. Purpose and scope

1.1 What this system is

ArabGN-CF is a measurement instrument, not a bias-detection product. It exists to make a specific class of claim checkable: that in Arabic, obligatory grammatical gender agreement carries the applicant's gender through any name-redaction safeguard, and that counterfactual hiring audits which claim "no bias" are usually underpowered to license that claim.

The system produces four deliverables:

ID	Deliverable	Type
C1	Cue-level prevalence of applicant-referring gender marking in Arabic recruitment text	Empirical finding
C2	ArabGN-CF — the open, frozen audit instrument	Software artifact
C3	Variance components and power/feasibility result for twin-contrast audits	Empirical + analytical
C4	Public pre-registration of the confirmatory study, timestamped before unblinding	Registered protocol

1.2 What this system is explicitly not

- It is not an audit result. Confirmatory twin contrasts stay sealed.
- It is not an evaluation of any commercial hiring system.
- It does not produce population statistics about real applicants. Twins are synthetic by necessity — no two real applicants have identical qualifications.
- Nothing built here is itself an audit subject.

1.3 Scope boundaries

- Language variety: Modern Standard Arabic. Dialectal CVs out of scope.
- Gender: treated as binary, because Arabic agreement morphology is binary. The agreement-free register is the closest the design comes to a neutral form, and its adequacy is itself a measured quantity, not an assumption.
- Audit subjects: frozen, version-pinned, open-weight systems only. Measured, never trained.

2. System overview

Five layers. Data flows top to bottom; the freeze layer gates everything below it.

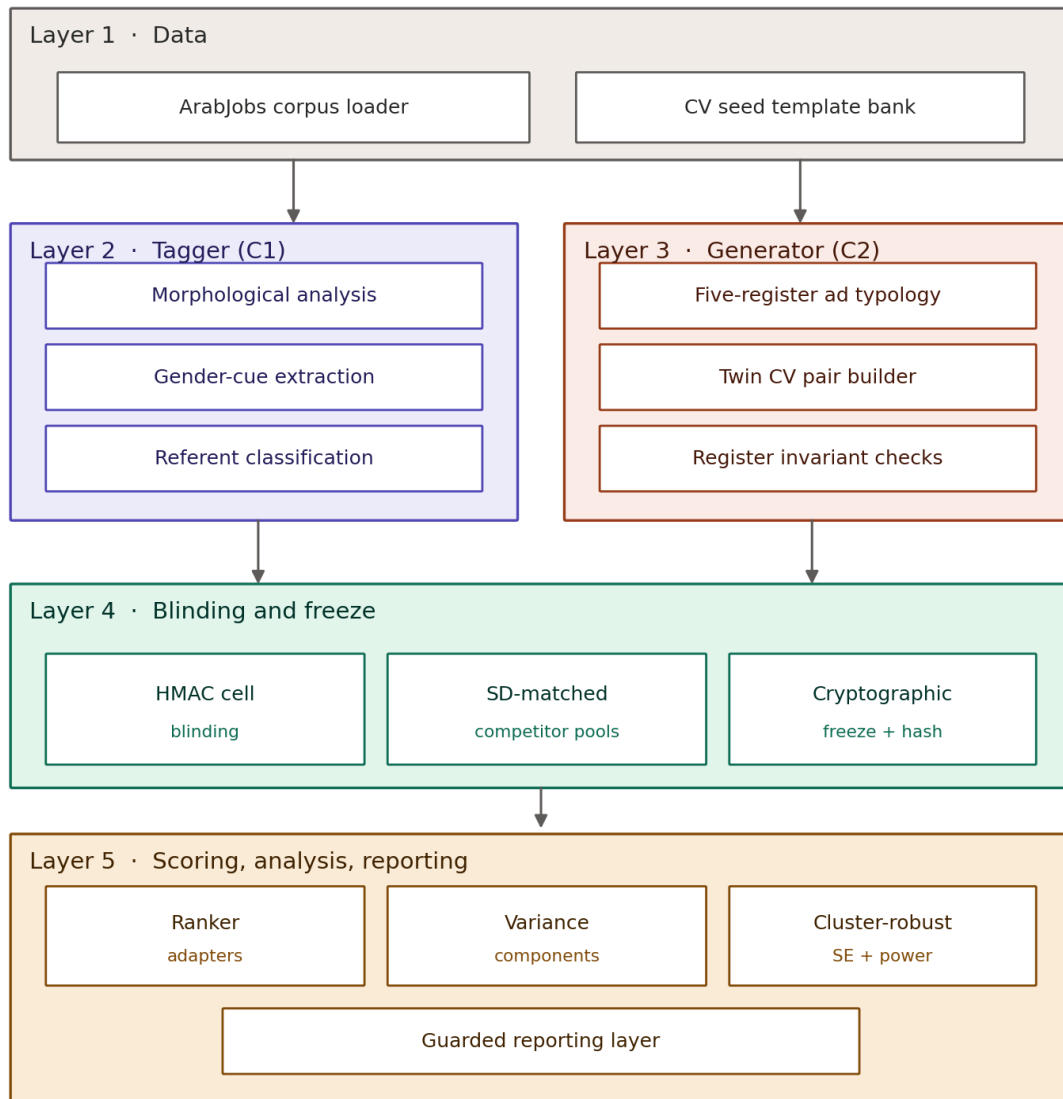


Figure 1 — Five-layer system architecture.

Two scoring backends exist behind one interface:

- Deterministic synthetic backend — pure function, no model downloads, runs on a laptop. Lets a reviewer re-run every estimator against known ground truth. This is what the 117 tests exercise.
- Real ranker backend — the version-pinned open-weight audit subjects.

3. Layer 1 — Data

3.1 Inputs

Input	Source	Licence	Status
ArabJobs corpus	arXiv:2509.22589	CC-BY	Redistribution query outstanding
CV seed templates	To be assembled	To be determined	Not started
Occupation/seniority taxonomy	Derived from ArabJobs metadata	—	Not started
Run configuration	Authored in-repo	—	Not started

3.2 Corpus loader — contract

Input: raw ArabJobs distribution. Output: a normalised record per advertisement.

```
AdRecord
  ad_id      str    stable hash of source text
  text_raw   str    unmodified source
  text_norm  str    Unicode-normalised, diacritics preserved
  country    enum   EG | JO | SA | AE
  occupation str    from source metadata
  seniority  enum   entry | mid | senior | unspecified
  source_checksum str
```

Normalisation rules (must be frozen and tested): Unicode NFC; alef/ya/ta-marbuta normalisation is **not** applied — ta-marbuta is the primary feminine marker and normalising it would destroy the signal being measured. This is the single most important preprocessing decision in the system and needs an explicit test asserting ta-marbuta survives.

4. Layer 2 — Tagger (C1)

4.1 Toolkit decision

CAMEL Tools 1.6 with morphology-db-msa-r13, verified 12 August 2026.

The database exposes three separate fields that the design depends on:

Field	Meaning	Why it matters
gen	Functional gender	The semantically real gender
form_gen	Surface / form gender	Diverges on broken plurals
rat	Rationality (humanness): r / i / n	Distinguishes person-referring from thing-referring

The rat field is what does the applicant-vs-job-property work the proposal describes. This is not a heuristic we invented — it is an established feature in the Arabic agreement-modelling literature (Alkuhlani & Habash) with published annotation guidelines, which strengthens C1’s defensibility.

4.2 Verified behaviour

Probed 12 August 2026 against the built-in MSA database:

Token	rat values returned	Interpretation
المهندسة	r only	Unambiguously applicant-referring
خبرة / الشركة / سنوات	i only	Unambiguously not the applicant
حاصلة	r and i	Genuine ambiguity
واسعة (adjective)	n	Inherits rationality from head noun
تخرجت (verb)	n	Inherits rationality from subject

Also observed: feminine surface forms frequently return both masculine and feminine analyses out of context. Out-of-context analysis is therefore insufficient — the disambiguator must run over full sentences.

4.3 Three-tier referent classifier

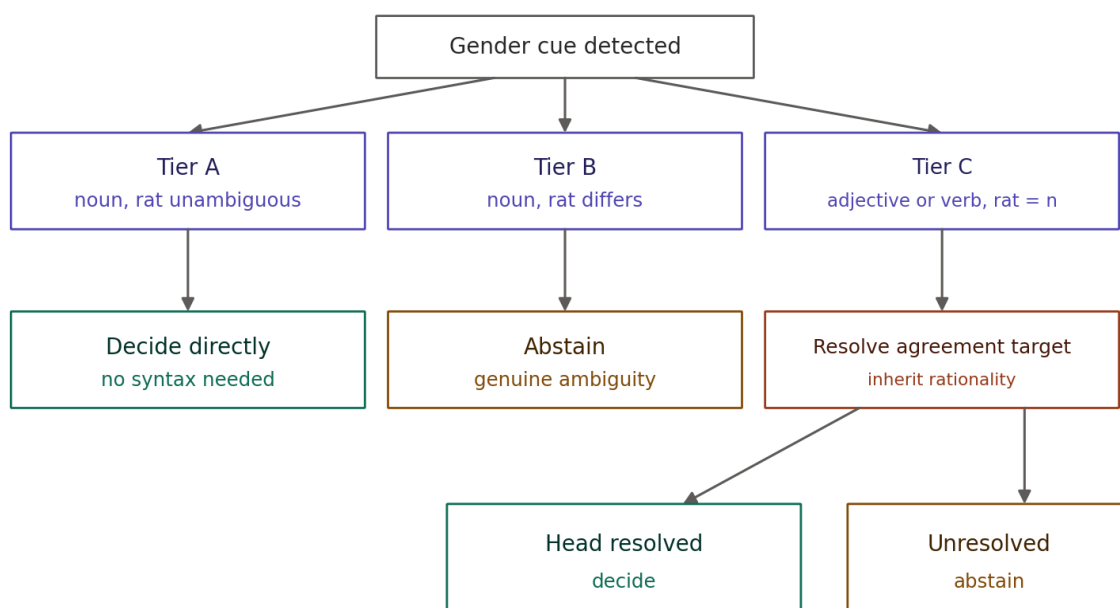


Figure 2 — Referent classification and abstain routing.

Tier C is the core of C1, not a refinement. The highest-value applicant cues — تخرجت, حاصلة, مطلوبة — are verbs and participles carrying rat = n. A tagger with only Tiers A and B will miss most of what the paper is about.

4.4 Abstain policy

Abstain fires on any of:

- Rationality disagreement — candidate analyses disagree on rat (Tier B).
- Unresolved agreement target — Tier C, no head noun recoverable.
- Gender disagreement — candidate analyses disagree on gen after disambiguation.
- Form/functional divergence — gen $\neq$ form_gen, flagged for review.

The first trigger is derived from the morphology database rather than a tuned threshold, which makes it defensible in the paper without calibration data.

4.5 Output contract

TaggedCue		
cue_id	str	
ad_id	str	
token	str	
char_span	(int, int)	
sentence_context	str	
pos	str	
gen	enum	m f
form_gen	enum	m f
rat_candidates	set	subset of {r, i, n}
tier	enum	A B C
referent	enum	applicant non_applicant ABSTAIN
abstain_reason	enum	null
head_token	str null	Tier C only
toolkit_version	str	
db_version	str	

4.6 Known limitation to report

مطلوبة returns rat = i on all readings, but in recruitment ads it frequently does refer to the sought person ("مطلوبة مهندسة"). The morphology database encodes general MSA rationality, not recruitment register. Expect a systematic error class here. It should be over-sampled in adjudication and reported as a named limitation rather than discovered by a reviewer.

5. Layer 3 — Generator (C2)

5.1 Five-register ad typology

Register	Description	Machine-checked invariant
R1	As-found generic masculine	Contains ≥1 applicant-referring masculine cue
R2	Dual / inclusive	Both gendered forms present for each applicant cue
R3	Agreement-free	Zero applicant-referring gender cues (verified by the tagger)
R4	Syntax-matched masculine placebo	Matches R2 in length and structure, no inclusive semantics
R5	Cross-lingual English	No grammatical gender agreement

R3’s invariant is enforced by the Layer 2 tagger, which creates a dependency: the tagger must be trustworthy before the generator can certify a register. This is a second reason to build Layer 2 first.

R4 exists to separate "the model responds to inclusive framing" from "the model responds to longer or more complex text."

5.2 Twin CV pair generator

Produces (female, male) CV pairs identical in every respect except gender agreement morphology.

Revised matching invariant. The proposal specifies refusal to emit pairs differing in token count. This is likely unsatisfiable: حاصل and حاصلة differ under every subword tokenizer in the audit set, so a strict equality constraint either blocks all output or forces silent padding, which is itself a confound.

Replace with a set of invariants that are satisfiable and honest:

- Identical content-word count.
- Identical qualification slot values (degree, years, institution, skills).
- Character-length difference within a declared tolerance.
- Per-pair tokenization delta measured and reported per audit subject, not forced to zero.
- Zero difference in any non-gender lexical item, asserted by diff.

This converts an unenforceable guarantee into a measured covariate, which is a stronger position under review.

5.3 Generation method — open decision

Method	Control	Naturalness	Leakage risk
Template-based	Highest	Lowest	Lowest
LLM-generated under constraints	Lowest	Highest	Highest — style may encode gender beyond morphology
Hybrid: template skeleton, LLM-filled slots	Medium	Medium	Medium, containable

Recommendation pending: the hybrid, with LLM output passed through the Layer 2 tagger to verify no unintended gender cues entered the "gender-neutral" content slots. Note that any LLM in the generator must be version-pinned and included in the freeze.

6. Layer 4 — Blinding and freeze

6.1 HMAC cell blinding

Cell identity (which register, which twin polarity) is hidden from everyone who prepares or scores material. Requirements:

- Key held outside the analysis repository.
- Blinding applied before material reaches the scoring layer.
- Unblinding is a separate, logged, one-way operation.
- Tests must confirm no cell identity leaks through file ordering, filename, or record ordering — ordering leaks are the most common failure mode here.

6.2 SD-matched competitor pools

Each twin is scored against a pool of competitor CVs matched on score standard deviation, adopted as method from Goyal et al. (arXiv:2604.06097). Pool composition is drawn once, frozen, and reused across cells so that pool variation does not enter the twin contrast.

6.3 Cryptographic freeze

Hash over: run config, every analysis module's source, corpus checksums, model version pins, and dependency lockfile. Confirmatory analysis refuses to run if the hash does not match.

Design change required. A hash you compute and print in your own paper proves the config did not drift. It does not prove the analysis predates unblinding, because you control both the artifact and the clock. It needs an external time anchor:

- OSF Registries entry (gives a timestamped DOI, and matches the social-science pre-registration norm the paper invokes), or
- AsPredicted, or
- An OpenTimestamps proof anchored to a public blockchain.

Cost is roughly an hour. Without it, C4's central claim is not independently verifiable.

7. Layer 5 — Scoring, analysis, reporting

7.1 Audit subjects

Subject	Type	Pin
multilingual-e5-large	Embedding ranker	Exact revision hash
BGE-M3	Embedding ranker	Exact revision hash
jina-v3	Embedding ranker	Exact revision hash
Arabic encoder (AraBERT / CAMELBERT / MARBERT — to select)	Embedding ranker	Exact revision hash
Open-weight LLM screener × 2	Generative screener	Exact revision + decoding params

Ecological validity note for the paper: bare cosine similarity is not how production ATS systems rank. The LLM screeners partly close this gap; the limitation should be stated rather than left for a reviewer to raise.

7.2 Outcome measures

Score level

- Per-pair score difference, female minus male.
- Mean difference per cell with cluster-robust interval.
- Per-cell score variance (adopted from Goyal et al.).

Selection level

- Selection rate at top-k of n (design point: top-10 of 100).
- Adverse impact ratio = SR_{female} / SR_{male} , benchmarked against the four-fifths rule, 29 CFR 1607.4(D).
- Rank displacement distribution.

7.3 Statistical machinery

Variance decomposition for a twin contrast over A ads × C CV twins:

$$SE^2 = \sigma^2_{ad} / A + \sigma^2_{cv} / C + \sigma^2_{resid} / (A \cdot C)$$

Open issue requiring resolution before the pre-registration is frozen. In a paired design where the female and male twin from the same pair are differenced, the pair-level random effect cancels in that difference and should not contribute to the SE of the contrast. If σ^2_{cv} here denotes twin-discordance variance — the variance of the difference across pairs — the formula is correct, but the notation reads like a CV-level random effect. A statistics reviewer will query it. Resolve and define explicitly in one sentence.

Required components

- Two-way cluster-robust variance over ads and twins. The proposal reports a 1.6× inflation on real structure versus treating resample draws as units, and that resample-as-unit understates SE by more than an order of magnitude — that comparison is itself a reportable finding.
- Holm correction. The family must be declared explicitly: subjects × registers × outcome measures multiplies quickly, and an undeclared family is a common reviewer objection.
- TOST or equivalent for the equivalence margins.
- Power curves over A and C at the declared margins.

7.4 Guarded reporting layer

Hard constraints, each with a test:

- Refuses to emit the phrase "no bias" in any output.
- Every interval reported with its clustering structure named.
- Equivalence claims blocked unless the achieved margin is met.
- Any cell with achieved power below the pre-registered floor is labelled inconclusive, not null.

8. Ground truth and evaluation metrics

There is no labelled "biased / not biased" ground truth, and the paper should say so directly. Four distinct validation sources substitute for it, and they answer different questions.

8.1 Tagger gold standard — the only direct ground truth

Human-adjudicated cue labels.

Metric	Definition	Target
Precision (applicant-referring)	$TP / (TP + FP)$	Report, do not pre-commit
Recall	$TP / (TP + FN)$	Report
F1	Harmonic mean	Report
Abstention rate	Abstains / total cues	Report; high is acceptable if precision on non-abstained is high
Inter-annotator agreement	Cohen's κ on double-annotated subset	≥ 0.7 before adjudication is usable
Tier-wise breakdown	All above, split by Tier A / B / C	Required — Tier C will be weakest and hiding that is not defensible

Sampling plan. Stratify by country, seniority, POS class, and tier. Over-sample abstentions and the مطلوبة error class. Double-annotate a subset for K.

8.2 Design-enforced invariants — not observed, asserted

Twin pairs differ only in gender morphology. This is guaranteed by construction and verified by machine-checked diff, not measured empirically. State this distinction plainly — it is a design guarantee, not evidence.

8.3 Positive control — instrument sensitivity

A cell with a deliberately injected, known-magnitude effect. If the instrument fails to recover it, every null elsewhere is uninterpretable. This is what makes a null result meaningful at all.

8.4 Zero-cue calibration cell — instrument specificity

A cell with no gender cue present. The instrument should return null. If it returns a signal, the pipeline has a leak — most likely through ordering, pool composition, or blinding failure.

Together, 8.3 and 8.4 are the instrument's ground truth. They deserve explicit framing in the paper as such, because they are the answer to "how do you validate a bias measurement with no bias labels."

8.5 C1 prevalence metrics

- Proportion of ads containing ≥ 1 applicant-referring gender cue, by country, seniority, occupation.
- Mean cues per ad, and per 100 tokens.
- Distribution over POS class and over tier.
- Distribution over register type as found in the wild.
- All of the above with abstentions reported separately, never silently dropped or silently counted.

8.6 C3 feasibility metrics

- Estimated σ^2_{ad} , σ^2_{cv} , σ^2_{resid} from the 40×50 pilot.
- Required A and C for each candidate margin.
- Cost translation: expert ad-authoring hours.
- SE inflation factor, resample-as-unit versus two-way cluster-robust.
- Attainability verdict per margin: the 1 pp principled margin versus the four-fifths benchmark.

9. Recommended build order

Ordered by dependency, with the human-lead-time item pulled forward.

Step	Work	Why here
1	Repository skeleton and data contracts	Cheap; every later module compiles against it. Includes the ta-marbuta preservation test.
2	Tier A/B cue extractor	Smallest piece producing real output. Testable against fixtures with no corpus, no adjudication, no models.

Step	Work	Why here
3	Adjudication tooling and gold set — start in parallel with Step 2	Only component with human lead time. All of §8.1 blocks on it and it cannot be compressed at the end.
4	Tier C dependency layer	Highest technical risk in C1. Deferred until Tiers A/B are validated so failures are attributable.
5	Full ArabJobs sweep → C1 tables	Mechanical once Steps 1–4 hold. Produces the first publishable finding.
6	Generator and register invariants	Depends on a trustworthy tagger for R3 certification.
7	Blinding, freeze, external time anchor	—
8	Analysis layer, pilot, power curves → C3	—
9	Pre-registration → C4	Depends on C3's numbers.

Why start here. C1 is the only contribution that is a genuine empirical finding and is fully independent of the twin machinery. It can be completed, checked, and written up without the generator, the blinding layer, or any scoring run. If the project stalls anywhere, C1 still stands alone as a publishable result. C3 by contrast depends on the generator and a scoring pass, and C4 depends on C3's numbers.

Start with Steps 1 and 2, and open Step 3 immediately alongside them.

10. Open decisions

#	Decision	Blocks
1	ArabJobs redistribution permission	Step 5 release, not Step 5 analysis
2	Twin CV generation method (§5.3)	Step 6
3	Which Arabic encoder as audit subject	Step 8
4	σ^2_{cv} definition in the SE formula (§7.3)	C4 freeze
5	Holm family declaration	C4 freeze
6	External time-anchor service (§6.3)	Step 7
7	CV seed template source and licence	Step 6
8	Whether C1's claim covers CVs or only ads (see §11)	Draft framing

11. Positioning issue to resolve before drafting

The introduction argues the gender marking sits inside the CV, where no employer edit and no redaction reaches. C1 measures prevalence in job ads. These are different corpora supporting different claims, and as written the motivating premise is not the thing C1 evidences.

Two resolutions:

- Cheap and honest: reframe C1 explicitly as measuring how ads refer to applicants, and let the CV claim rest on the grammar of Arabic rather than on a corpus measurement. Costs nothing, closes the gap.

- Stronger, more expensive: add a small hand-collected Arabic CV sample so the CV claim carries its own numbers.

This should be settled before drafting, not during.